Thyroid Eye Disease

Lorne Yudcovitch, OD, MS, FAAO

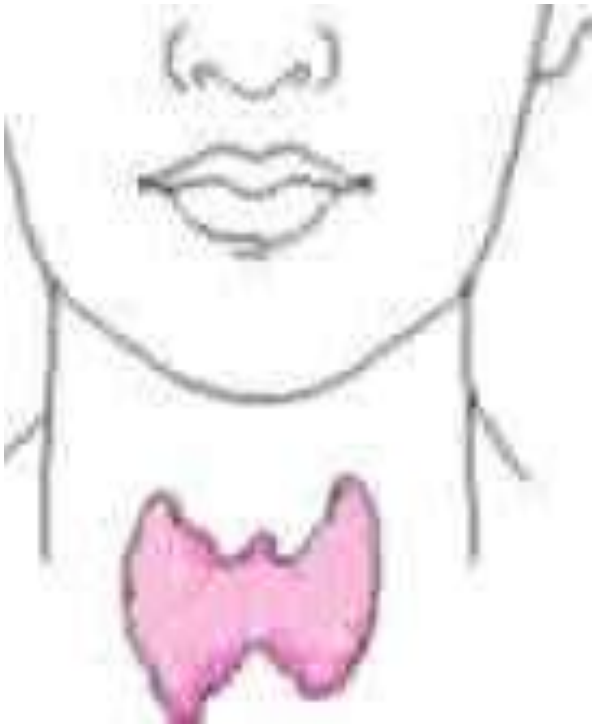
Lecture Acknowledgement and Thanks:

Professor William C. Hefner

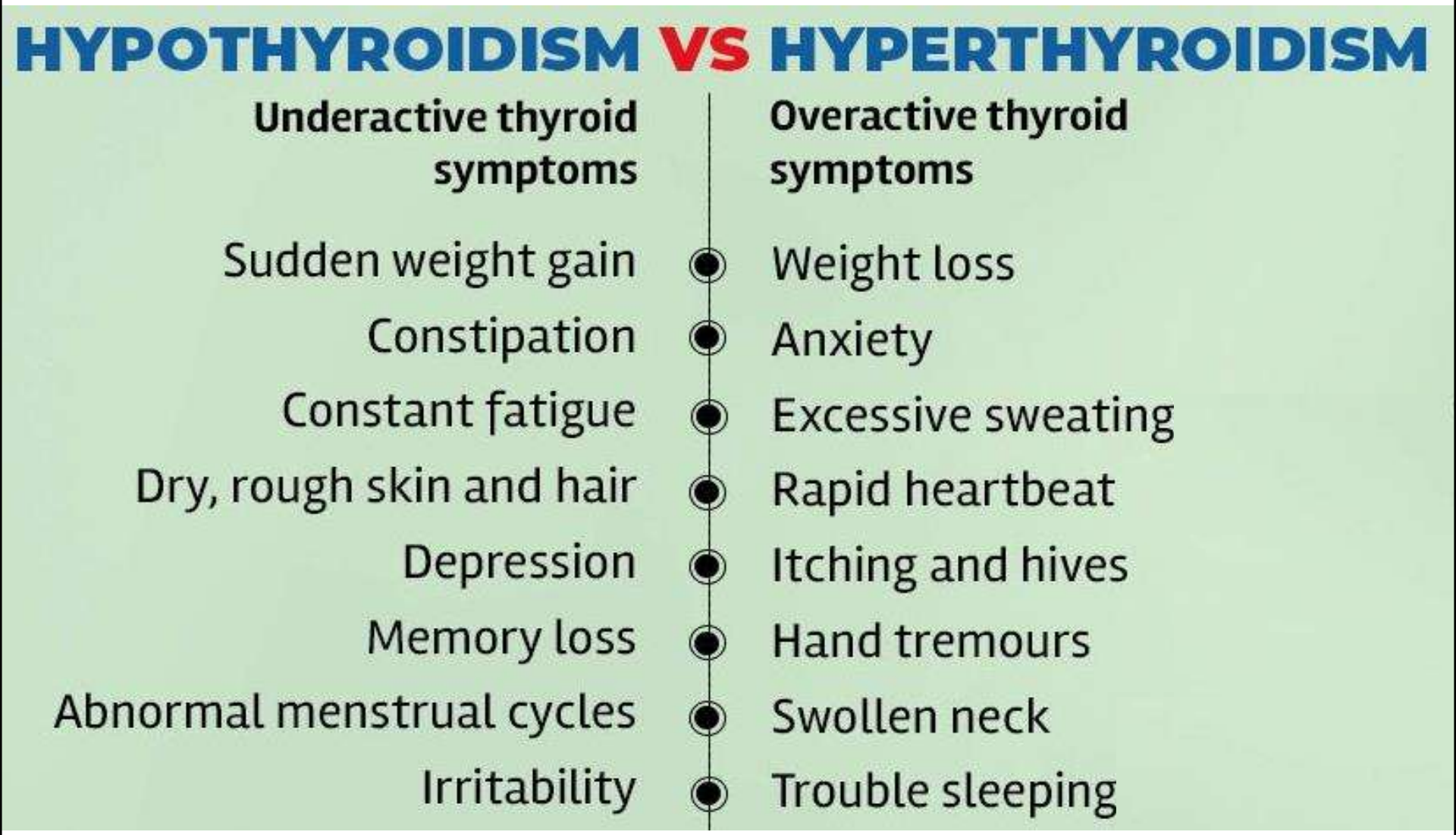
Pacific University



Thyroid Gland

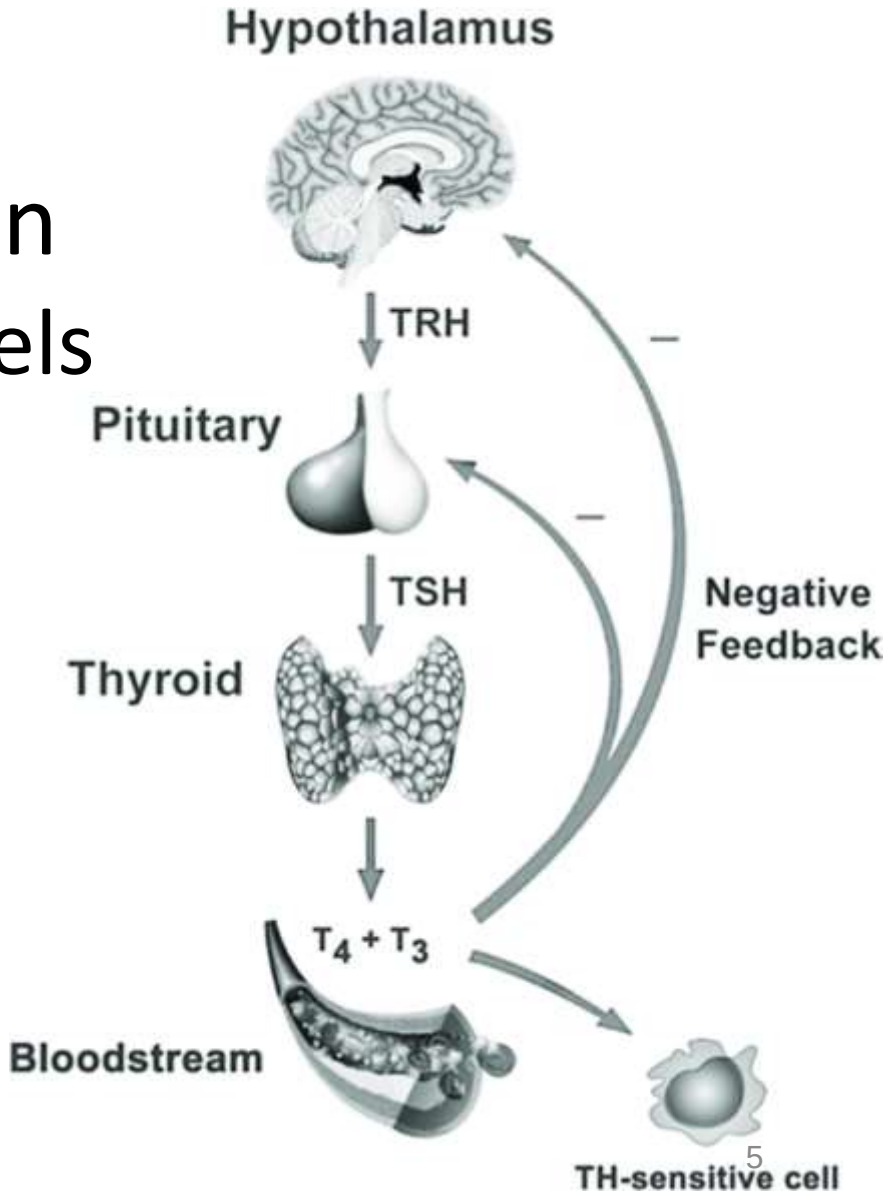


- Functions:
  - Growth
  - Metabolism
  - Development
- Conditions based on activity:
  - Euthyroid – subclinical dysfunction
  - Hypothyroid – under active
  - Hyperthyroid – over active


HYPOTHYROIDISM VS HYPERTHYROIDISM
Underactive thyroid symptoms
Overactive thyroid symptoms
Sudden weight gain
Weight loss
Constipation
Anxiety
Constant fatigue
Excessive sweating
Dry, rough skin and hair
Rapid heartbeat
Depression
Itching and hives
Memory loss
Hand tremours
Abnormal menstrual cycles
Swollen neck
Irritability
Trouble sleeping

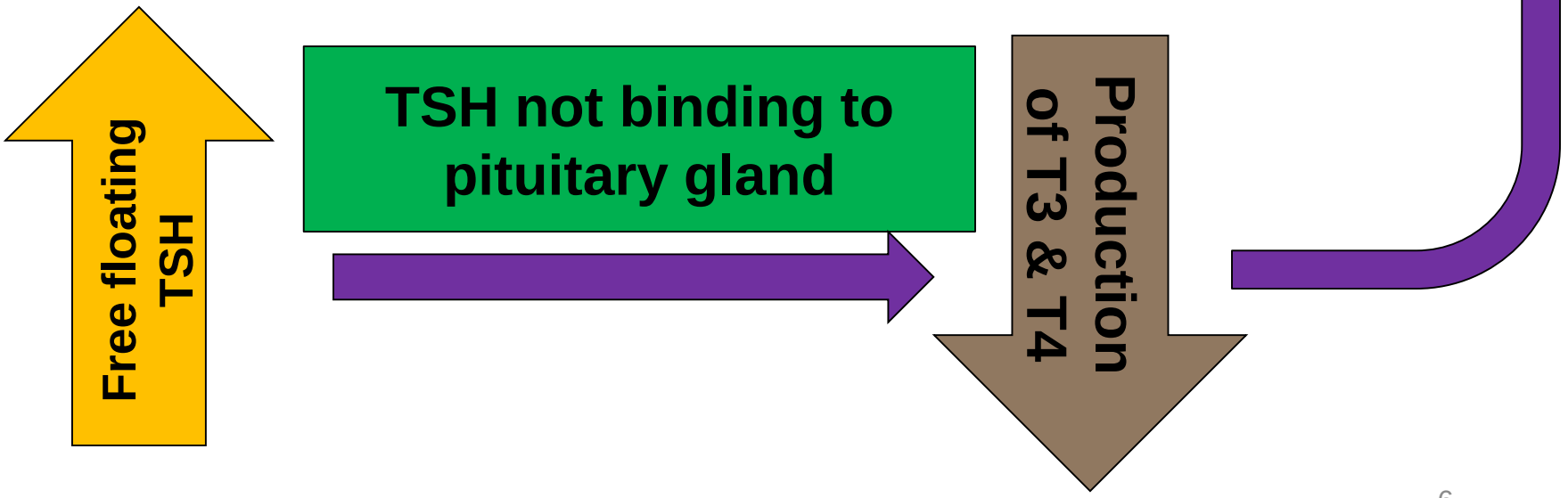
Euthyroid

- Subclinical Dysfunction
  - Normal T3 and T4 levels
  - Abnormal TSH levels
    - Elevated TSH → Hypothyroid
    - Depressed TSH→ Hyperthyroid



Hypothyroid

- Body lacks sufficient amounts of thyroid hormone
- ~5% in US have hypothyroidism
- Women >> Men by 5-7x
- Up to 10% of all women

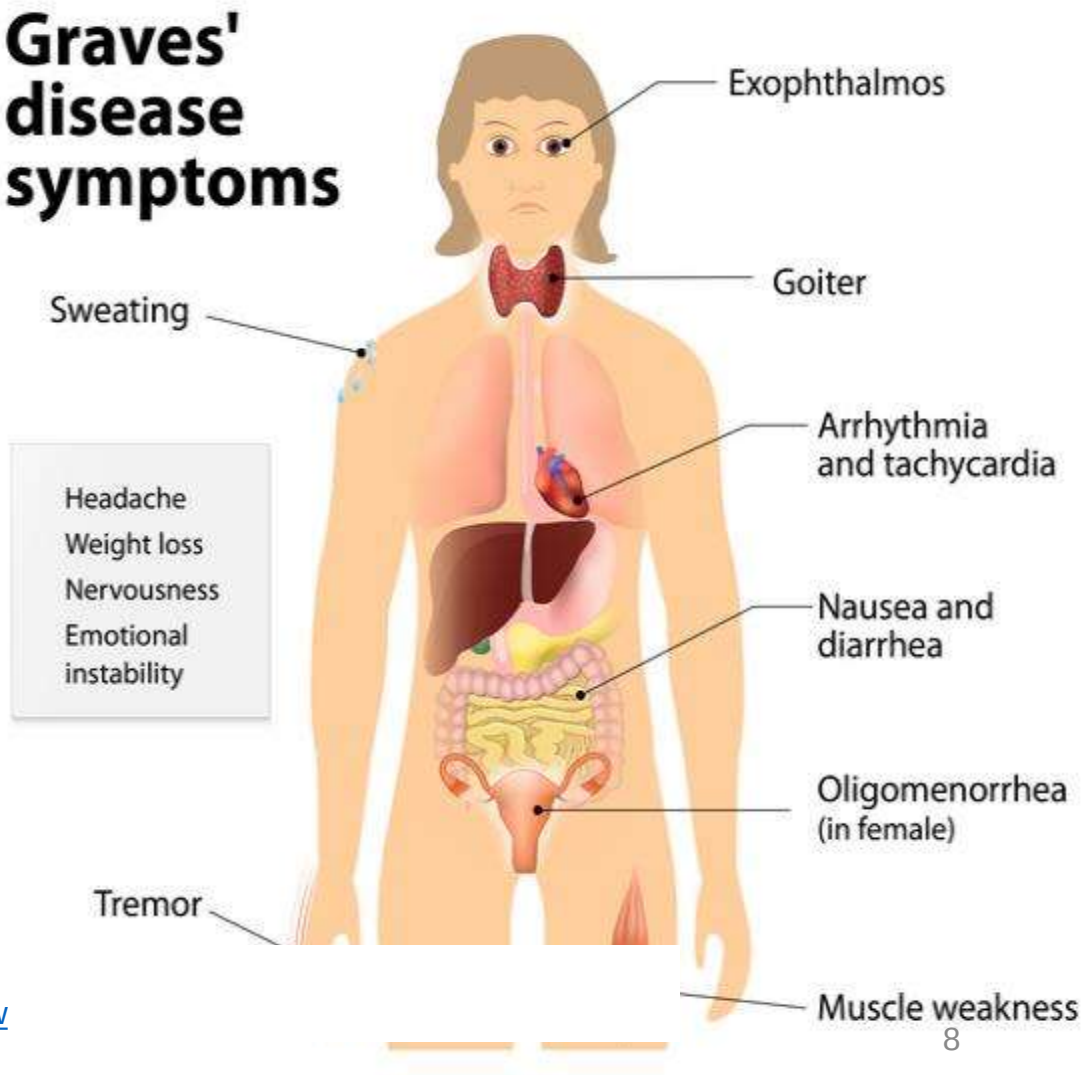


Hypothyroid

- Three major causes of hypothyroidism:
  1. Inflammation
     - Autoimmune thyroiditis (Hashimoto's Disease)
  2. Medical treatments
     - Partial/total removal of thyroid
     - Chemical or radiation destruction
  3. Medications
     - Lithium
     - Amiodarone

Hyperthyroid

- Too much thyroid hormone
- ~1% in US have it
- Most common cause: **Graves' disease**
  - T3>>T4



Sai-Ching JY, Khardor R. Graves Disease. http://emedicine.medscape.com/article/120619-overview

Points of Clarification

- Graves' Disease - systemic condition
- Graves' EYE Disease - ocular condition

Graves' Eye Disease "Aka's"	
Thyroid Eye Disease (TED)	Graves' Orbitopathy
Thyroid Associated Orbitopathy	Graves' Ophthalmopathy
Thyroid Ophthalmopathy	Dysthyroid Ophthalmopathy

Kanski's Clinical Ophthalmology. 2016. Ch 3.

Graves' Hyperthyroidism

- 13.9/100,000 incidence
  - Females >> Males (8:1)
  - Distribution doesn't vary with age
- Up to 50% patients may have orbitopathy
  - Severity varies with gender
    - Mild disease more likely in women
    - Men more likely to be severely affected
- Graves' Disease and TED run in parallel
  - Ocular issues can improve once thyroid is controlled

[1]Antonelli A, et al. *Best Pract Res Clin Endocrinol Metab.* 2020;34(1):101387
[2]From Duane' Ophthalmology – Chapter 36 – Thyroid Orbitopathy

Thyroid Eye Disease (TED)

- Multisystem autoimmune disorder
- 16 per 100,000 females
  - Usually ages 40–44 and 60–65 years
- 3 per 100,000 males
  - Usually ages 45 – 49 and 65 – 69 yrs
- Severe cases occur more often in men

Edsel I, Roy H. Thyroid-Associated Orbitopathy. http://emedicine.medscape.com/article/1218444-overview#showall
https://webeye.ophth.uiowa.edu/eyeforum/patients/thyroid-eye-disease.htm

TED Overview

- Thyroid Eye Disease w/euthyroid condition
  - Also known as Euthyroid orbitopathy
  - Pathogenesis not well understood
  - 5% cases
  - Recommend thyroid testing every 6 – 12 months

  - ***25% develop thyroid dysfunction in 1 year***
  - ***50% develop thyroid dysfunction in 5 years***

Edsel I, Roy H. Thyroid-Associated Orbitopathy. http://emedicine.medscape.com/article/1218444-overview#showall;
Duane's Ophthalomology – Chapter 36 – Thyroid Orbitopathy

TED Overview

- Risk Factors:
  - Smoking
    - THE major MODIFIABLE risk factor
    - 4x greater risk



Edsel I, Roy H. Thyroid-Associated Orbitopathy. http://emedicine.medscape.com/article/1218444-overview#showall

TED Overview

- Risk Factors:
  - Radioiodine
    - 60% of Tx for hyperthyroidism in US
    - Effectively ablates thyroid gland over 6-18 weeks
    - Somewhat controversial
    - Thionamide may be better option
    - May cause flare up of thyroid eye disease
  - Ethnicity
    - Black, Asian ancestry

Walsh, JP. Radioiodine and thyroid eye disease- use with caution. https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1116221/

Edsel I, Roy H. Thyroid-Associated Orbitopathy. http://emedicine.medscape.com/article/1218444-overview#showall

JAMA. 2014;311:1563-1565

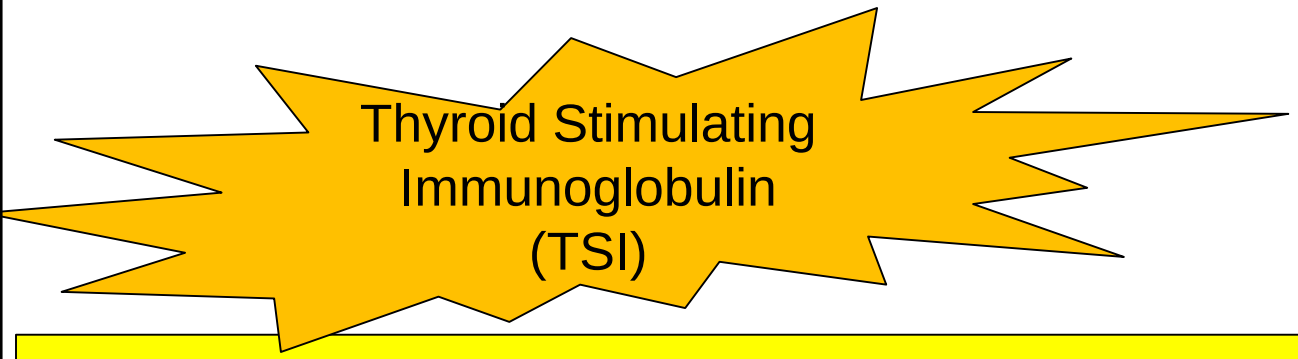
TED Mechanism

- Main autoantibody produced in TED is TSI
  - TSI = Thyroid Stimulating Immunoglobulin
  - Can be measured in blood to help monitor disease activity
  - Amount of TSI present correlates with TED severity

However, sometimes TED can occur without TSI formation
--think ‘euthyroid ophthalmopathy’--

https://webeye.ophth.uiowa.edu/eyeforum/patients/thyroid-eye-disease.htm

TED Mechanism Overview



Thyroid Stimulating Hormone Receptor (TSHR)
Insulin-Like Growth Factor-1R (IGF1R)

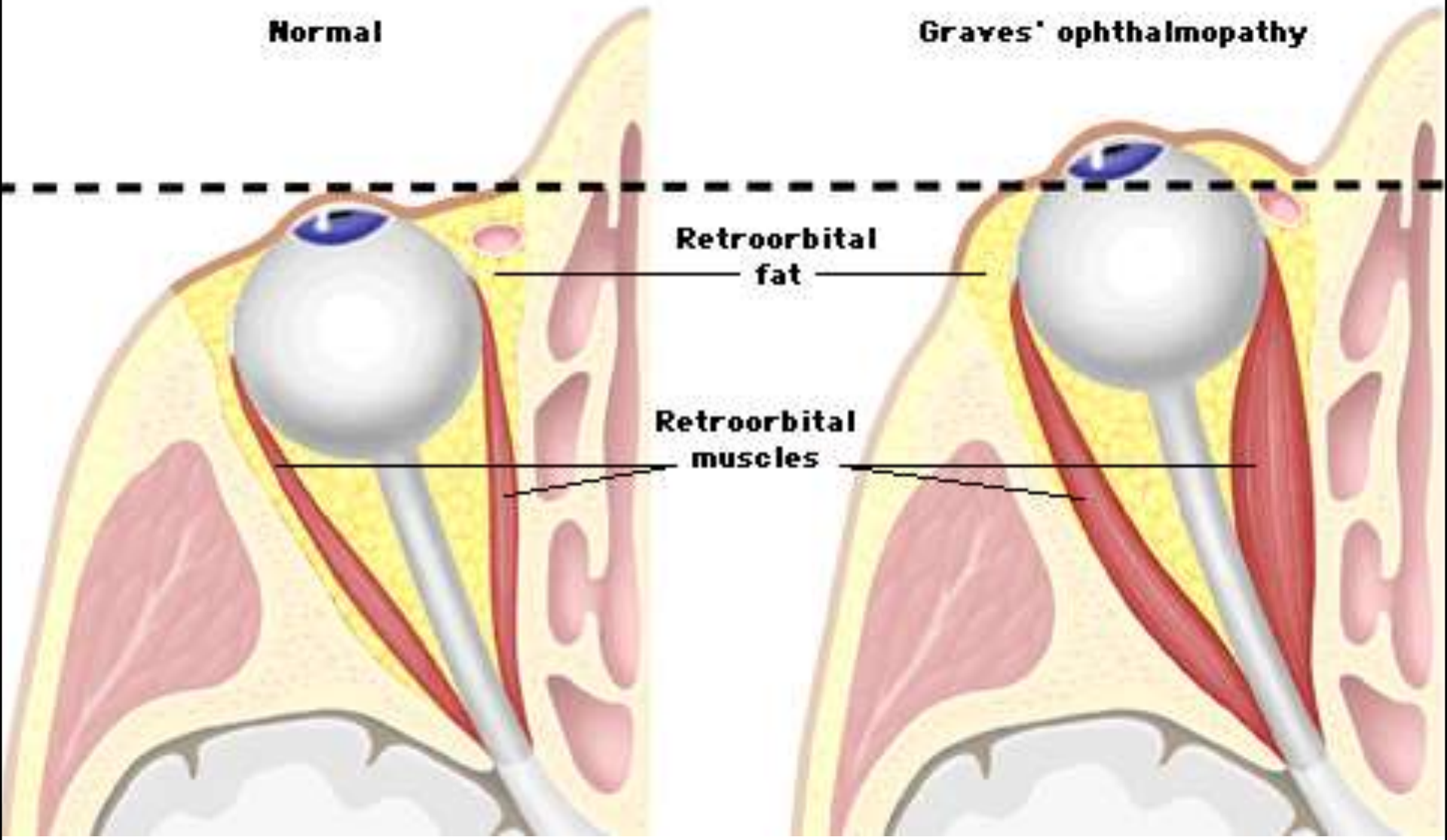
Fibroblasts stimulated

Inflammatory Cytokines

GaG's & Hyaluronic Acid

↑ Orbital Volume
- Proptosis
- Restricted EOM movement
- Optic Neuropathy

Proptosis Mechanism



http://www.thyroid.com.au/thyroid-disorders/thyroid-eye-disease/

EOM Mechanism

- Accumulation of GAG's, lipid, lymphocytes, and ultimately, fibrosis
- Enlarged ocular muscles and soft tissues
- Pushes eye forward → exophthalmos
- Eye movements reduced
- Strabismus
- IMSLO→

  Inferior, Medial, Superior,

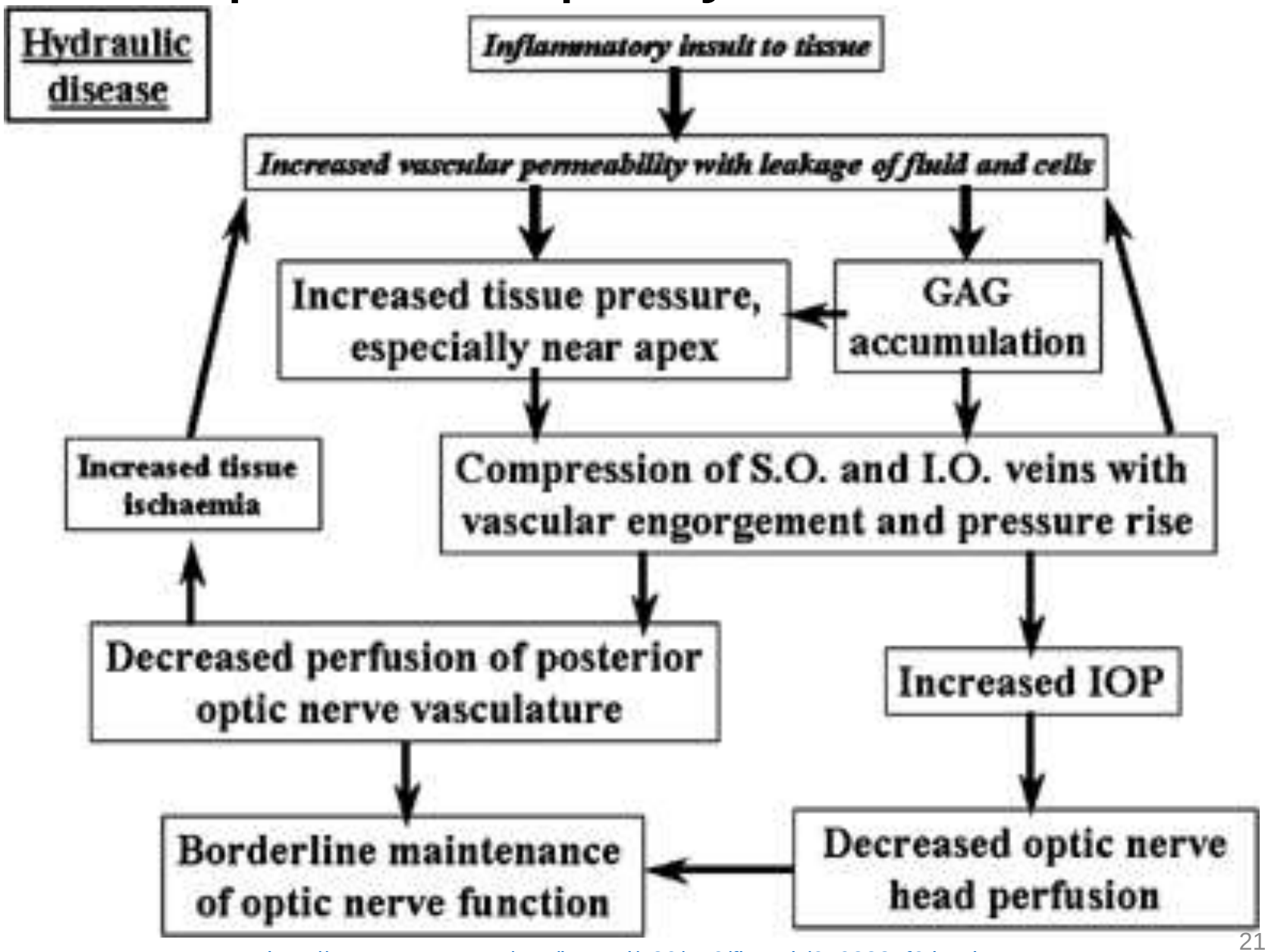
  Lateral, Oblique



http://www.learningradiology.com/caseofweek/caseoftheweekpix2/cow154lg.jpg

Adapted from www.medweb.bham.ac.uk

Optic Neuropathy Mechanism



http://www.nature.com/eye/journal/v20/n10/fig_tab/6702381f6.html

TED Clinical Course

Active Phase



Stable Phase



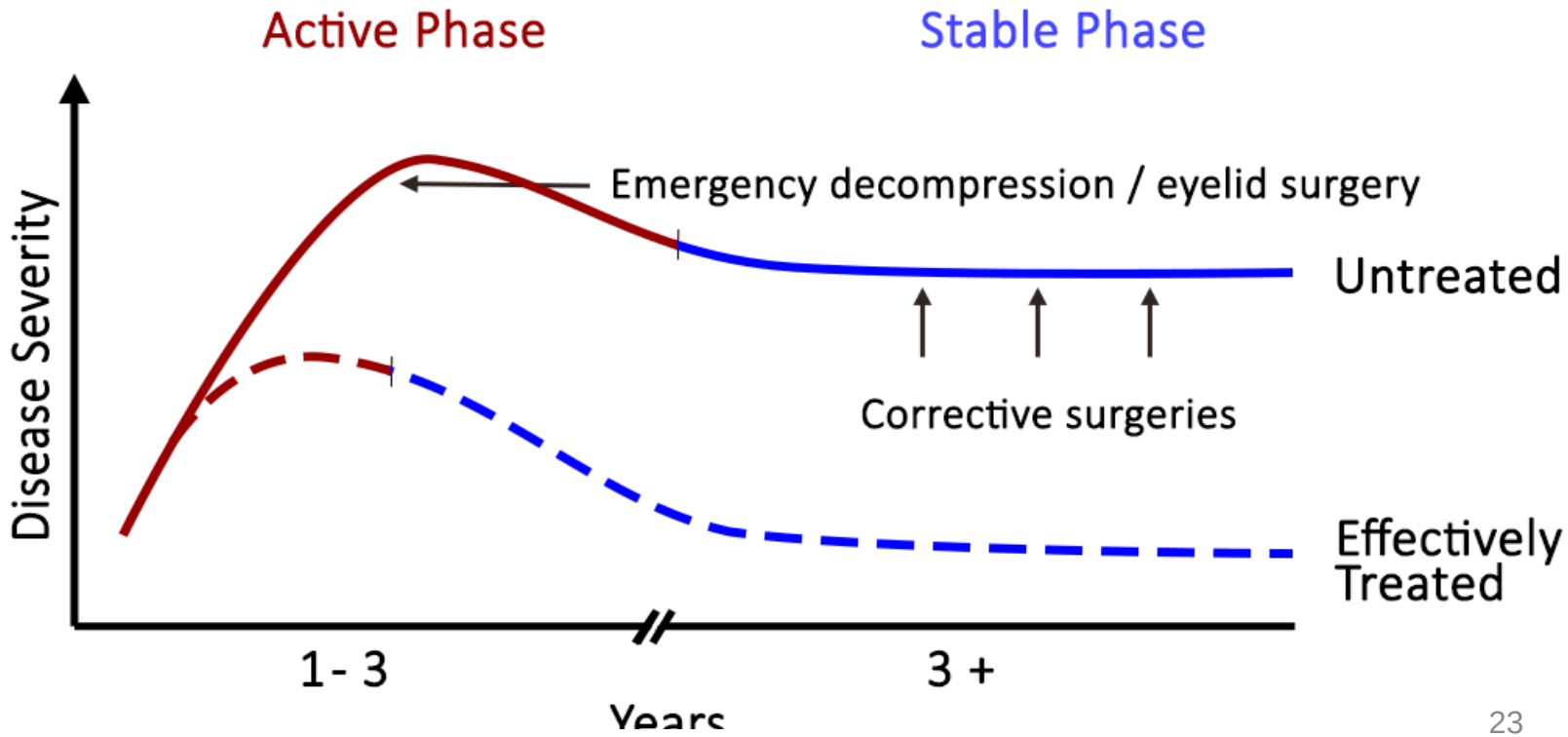
- Active Phase
  - = congestive phase = inflammatory
  - Red, painful eyes
  - Tends to remit within 1-3 years
  - Recurrence rate ~5-10%
- Stable Phase
  - TED less likely to recur after 18 months
- ~10% patients develop long-term ocular problems
- Start thyroid therapy → shorter active phase

Kanski's Clinical Ophthalmology. 2016. Ch 3.

TED Clinical Course

- Rundle's Curve – describes course of TED

Disease Time Course and Intervention Strategy



https://webeye.ophth.uiowa.edu/eyeforum/patients/thyroid-eye-disease.htm

TED Symptoms

- Dry Eyes
- Puffy Eyelids
- ‘Angry-looking’ eyes
- Bulging eyes
- Diplopia
- Visual acuity loss
- Visual field loss
- Dyschromatopsia
- Photopsia on upgaze
- Ocular pressure/pain

Edsel I, Roy H. Thyroid-Associated Orbitopathy. http://emedicine.medscape.com/article/1218444-overview#showall

TED Classifications

- Many types of classification
- All based on signs and symptoms
- We will discuss five:

1. By Structure / Clinical Manifestations
2. NOSPECS (mnemonic)
3. EUGOGO
4. Clinical Activity Score (CAS)
5. VISA

Classification by Structure

- Lids
- Globe
- Conjunctiva & Cornea
- Extraocular muscles (EOMs)
- Fundus
- Optic nerve head (ONH)

Classification by Structure

Lids

- Puffy lids
- Lid retraction - lid margin at/above superior limbus
  - Present 75% at time of diagnosis
  - 90% patients will ultimately manifest





Classification by Structure

Lids

- Lid lag
  - **Von Graefe's sign**: Delay of upper lid when following globe movement in down gaze
  - **Dalrymple sign**: Lid retraction in primary gaze
  - **Kocher's sign**: Globe lagging behind upper lid on up gaze
  - **Boston's sign**: Jerky downward movement of lid
  - **Griffith's sign**: Delay of lower lid in following globe movement in up gaze

Classification by Structure

Lids

- Other Lid signs
  - **Rosenbach's Sign**: Tremor of closed eyelids
  - **Stellwag's Sign**: Reduced blinking
  - **Jellinek's Sign**: Increased pigmentation of skin of eyelids

Classification by Structure

Lids

Medical Management

- Lubrication
- Topical alpha adrenergic blockers (relaxes muscles)
  - guanethidine, reserpine, thymoxamine
  - dapiprazole (REV-Eyes)
  - brimonidine 0.025% (Lumify)
- Topical beta adrenergic blockers
- Systemic beta adrenergic blockers (i.e. propranolol)
- botulinum A toxin (Botox)

Classification by Structure

Lids

Surgical Management

- Surgery - wait 6 months to 1 year!
  - Levator surgery
  - Muller's muscle recession

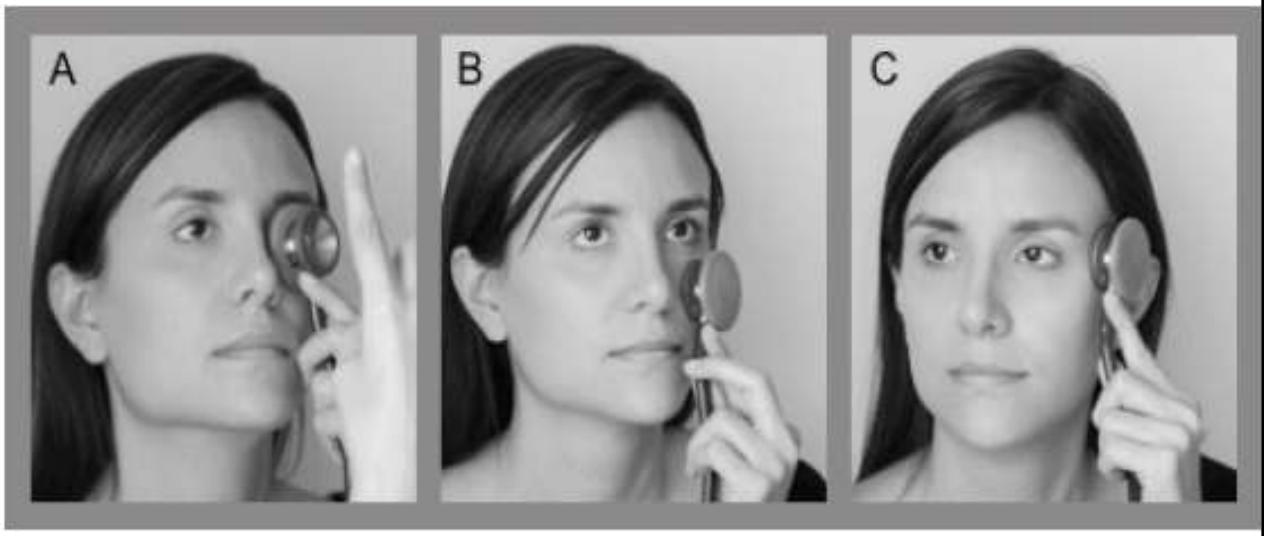
What is going on here?



Unilateral proptosis



Classification by Structure



Globe

- Ocular bruit
  - Upon auscultation, can hear increased blood flow through collateral arterial system
- Exophthalmos
  - 20-30% of Graves' disease patients
  - 40-70% of patients with TED
  - Bilateral in 80-90% of cases

Classification by Structure

- Proptosis / Exophthalmos
  - Measured with exophthalmometer



  - 2 mm or more above “normal” limit:
    - African Americans = 22mm
    - Caucasians = 20 mm
    - Japanese descent = 18 mm

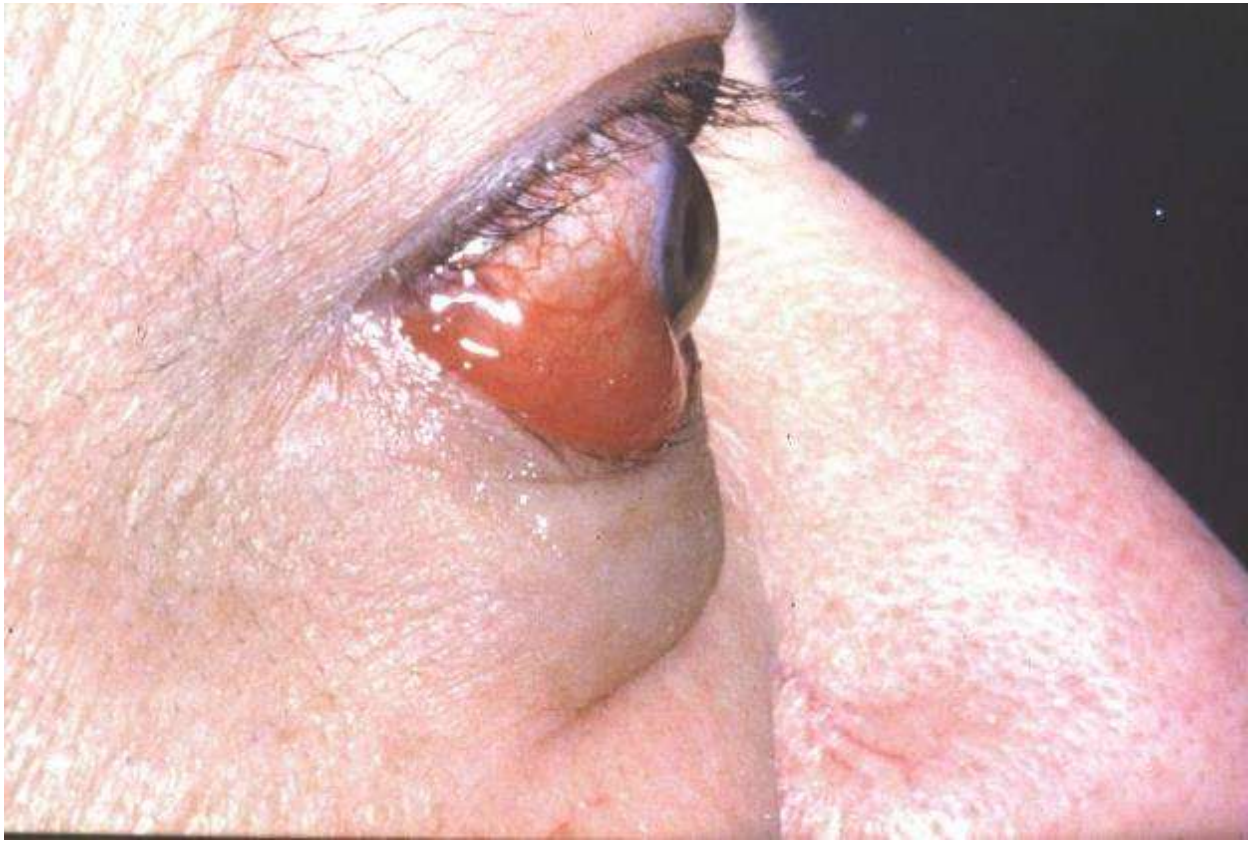


  - Retropulsion useful
    - Limited in those with severe thyroid eye disease

What is the problem here?



Conjunctival chemosis



Classification by Structure

- Conjunctiva and Cornea
  - Injected, chemotic conjunctiva
  - Superior limbic keratoconjunctivitis
  - Signs of corneal exposure and ulceration

Exposure keratopathy



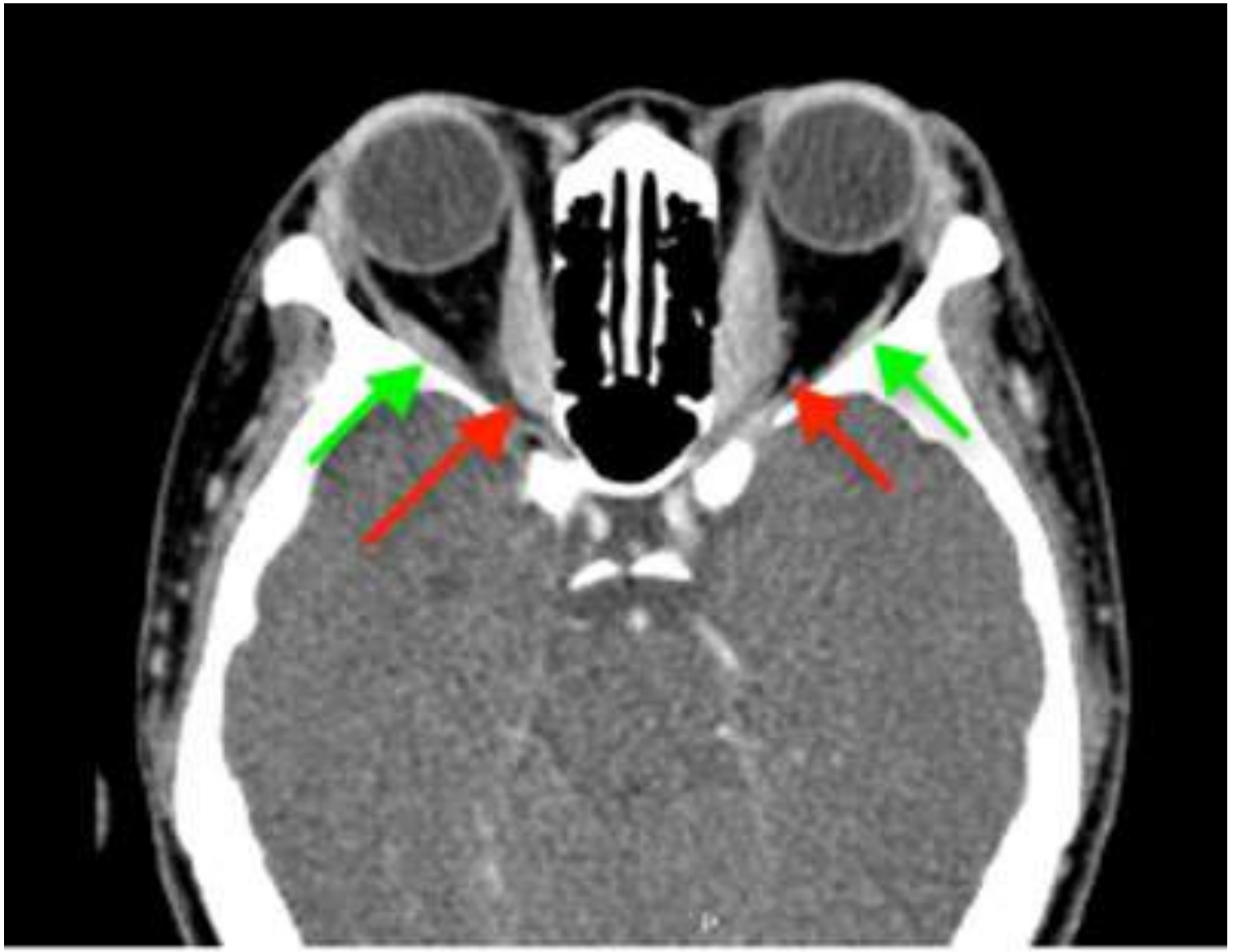
Classification by Structure



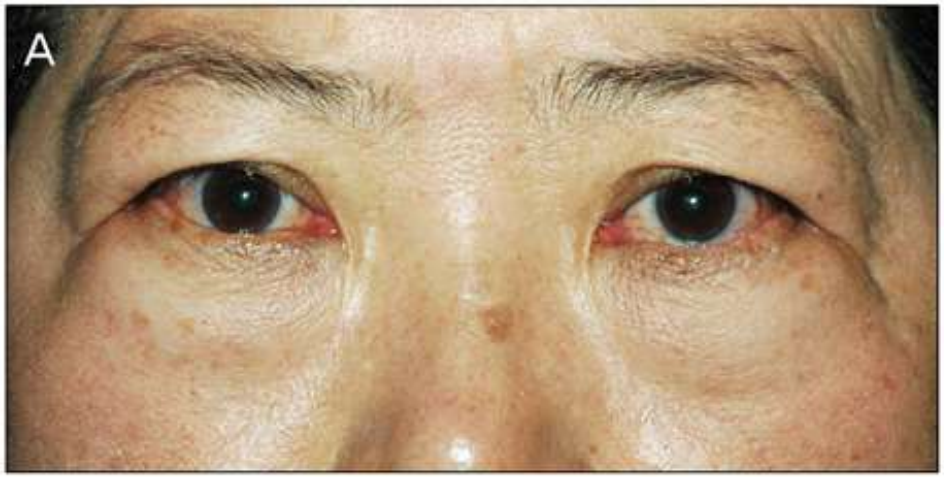
Conjunctiva & Cornea:

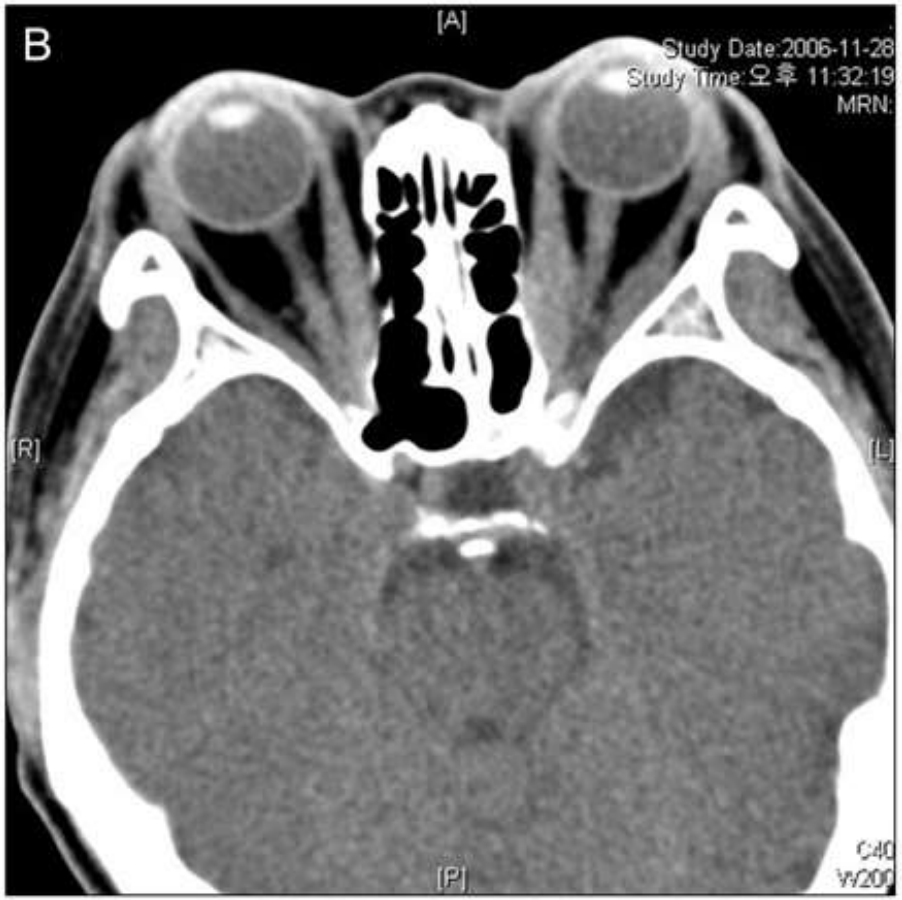
- **Medical Management**
  - Lubrication
  - Elevate head during sleep
  - Tinted cosmetic lenses
  - Steroids (systemic, topical, periocular)
  - Antibiotics
  - teprotumumab (TEPEZZA)
- **Sugical Management:**
  - Orbital Decompression

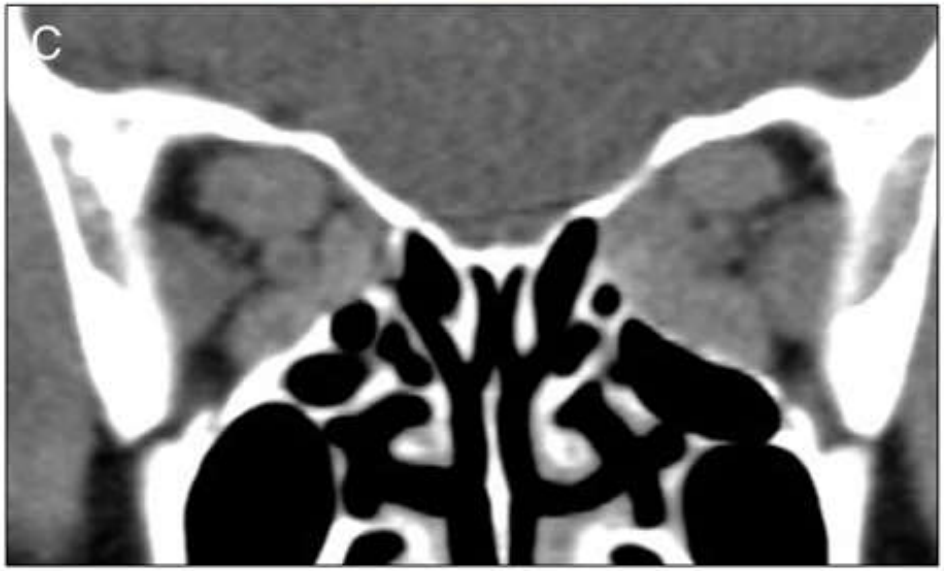
What is the problem here?



http://www.neec.com/Pages/Services/Neuro-Ophthalmology/thyroid-eye-disease.php







http://ekjo.org/search.php?where=aview&id=10.3341/kjo.2010.24.5.267&code=0065KJO&vmode=PUBREADER#!po=1.78571

Classification by Structure

Extraocular Muscles

- 60-98% pts w/Graves' Disease
  - Can swell to eight times their normal size!

- Restrictive extraocular myopathy in ~42% pts w/TED

Duane's Ophthalomology – Chapter 36 – Thyroid Orbitopathy
Kanski's Clinical Ophthalmology. 2016. Ch 3.

Classification by Structure

Extraocular Muscles

- Apparent under-activity of extraocular muscles
  - Most common: IR fibrosis
    - Causes diplopia on up-gaze
  - 2nd most common: MR
  - Then SR and Levator
  - Finally LR
- Elevated intraocular pressure on up gaze

What is the problem here?

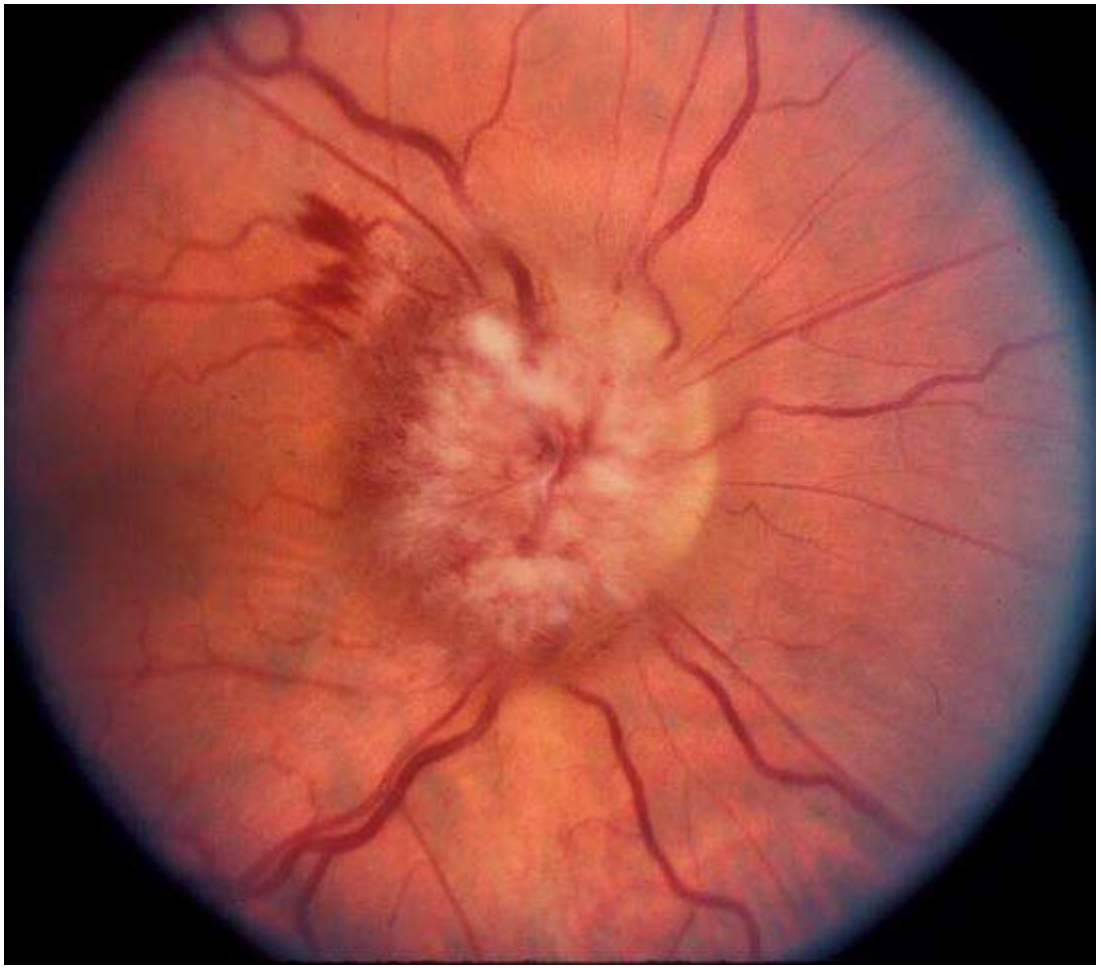


Classification by Structure

EOM's: Plan

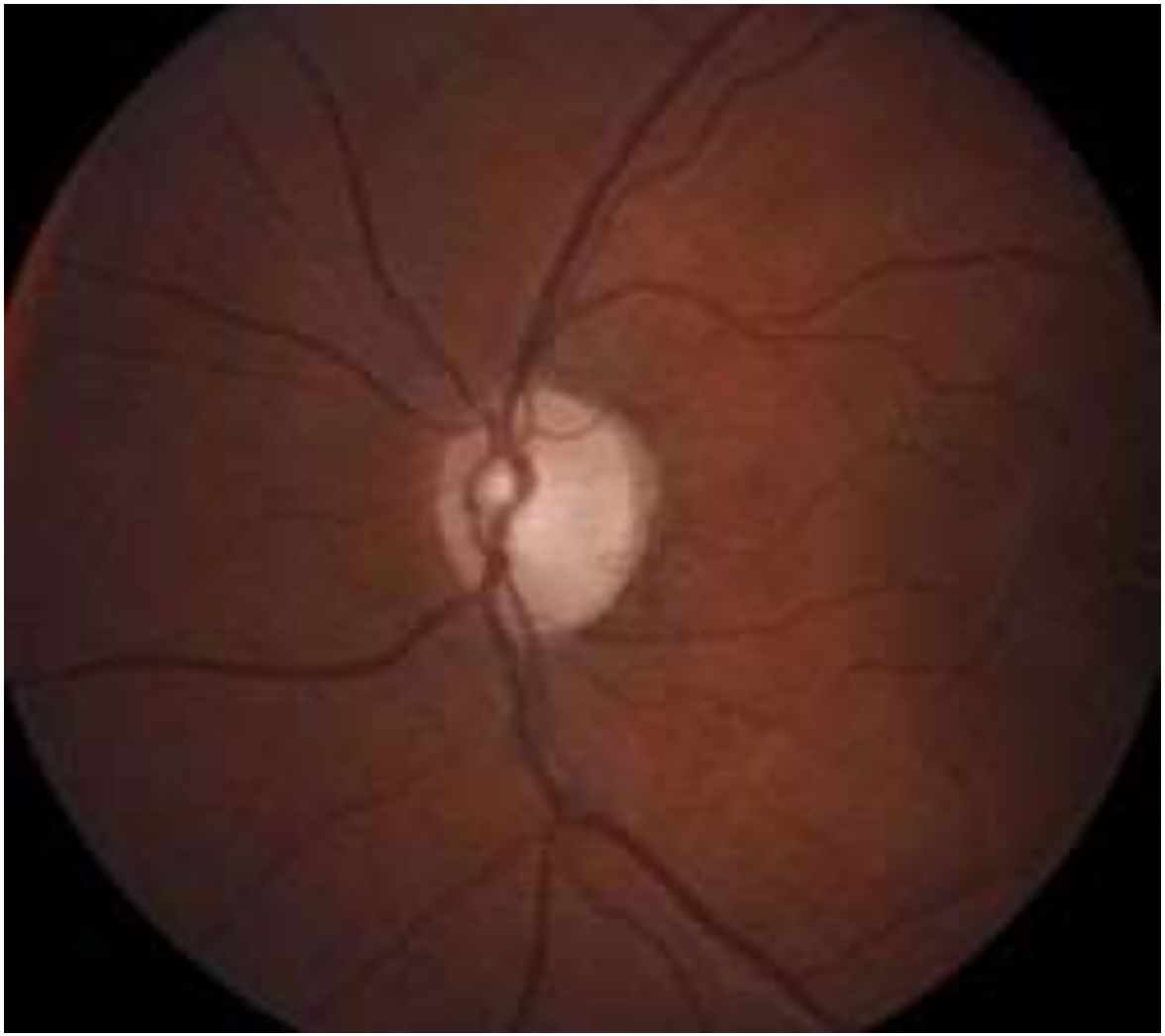
- **Medical Management**
  - Lubrication
  - Eye patching or prisms
  - Meds
    - Steroids
    - botulinum toxin
  - Surgery - wait 6 months to 1 year

What is the problem here?



https://www.merckmanuals.com/professional/eye-disorders/optic-nerve-disorders/papilledema

What is the problem here?

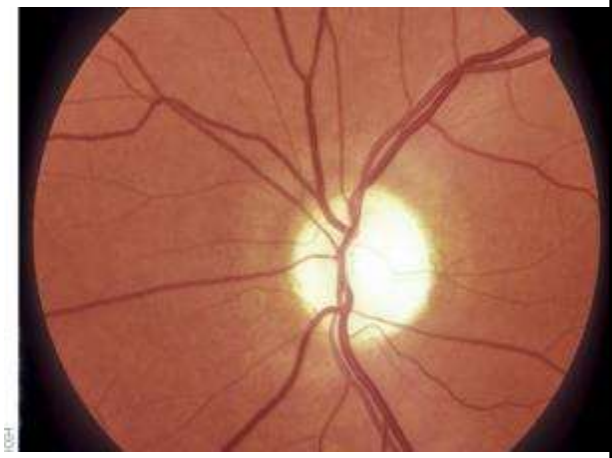


https://aapos.org/glossary/optic-nerve-atrophy

Classification by Structure

Fundus

- Optic neuropathy with vision loss <5%
- Optic neuropathy: most common blindness cause from TED
- Normally older (50-70 years old)
- Later onset thyroid disease
- More often have diabetes
- Normally bilateral, but up to 1/3 unilateral



Classification by Structure

Fundus

- Symptoms of Compressive Optic Neuropathy:
  - Blur
  - Vision loss
  - Color vision changes
  - Dyschromatopsia
  - Visual field loss
  - Headaches

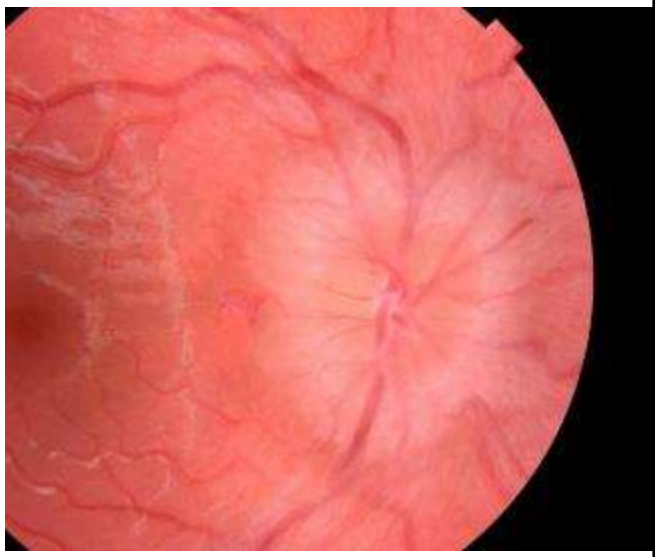
Classification by Structure

Fundus

- Compression of Optic Nerve results in…
  - Afferent Pupillary Defect in 35%
  - Abnormal (Pale/Swollen) optic disc 53%
  - Visual Field Defects 66%
  - May also see:
    - Congested retinal veins
    - Chorioretinal striae

Duane's Ophthalomology – Chapter 36 – Thyroid Orbitopathy

Classification by Structure



Optic Nerve

- Optic disc edema
- Painless, gradual vision loss
- 70% improve without TX!
- Management
  - Correction of thyroid imbalance
  - Meds
    - IV then oral Steroids
    - Cytotoxic or immunosuppressive agents
  - Orbital decompression
  - Orbital radiation

Classification- NOSPECS

NOSPECS

- Widely used to classify severity
- Limited because:
  - Inability to measure progression RATE
  - Inability to measure treatment-induced regression

Adapted from Duane's

Class	Definition
0	No physical signs or symptoms
1	Only signs, no symptoms (e.g. upper eyelid retraction, stare and eyelid lag)
2	Soft tissue involvement (symptoms and signs)
3	Proptosis
4	Extraocular muscle involvement
5	Corneal involvement
6	Sight loss (optic nerve involvement)

Each class usually, but not necessarily, includes the involvements indicated in the preceding class

From Werner SC. Modification of the classification of the eye changes of Graves' disease. Am J Ophthalmoi 1977;83:725–727, and ETA, LATS, Japanese-AOTA, ATA. Classification of eye changes of Graves' disease. Thyroid 1992;2: 235–236.

NOSPECS

Class 0 – No signs or symptoms

- Management:
  - Routine physical and eye exams



Marcocci, C., Smith, T.J. (2016). Graves' Ophthalmopathy. In: Vitti, P., Hegedus, L. (eds) Thyroid Diseases. Endocrinology. Springer, Cham. https://doi.org/10.1007/978-3-319-29195-6_16-1

NOSPECS

Class 1 – Only signs/no symptoms

- Non-infiltrative stage
- Upper eyelid retraction (Dalrymple sign)
  - May resolve spontaneously
- Upper lid "lag" on down gaze (von Graefe sign)
- Management:
  - Lubricants



Marcocci, C., Smith, T.J. (2016). Graves' Ophthalmopathy. In: Vitti, P., Hegedus, L. (eds) Thyroid Diseases. Endocrinology. Springer, Cham. https://doi.org/10.1007/978-3-319-29195-6_16-1

NOSPECS

Class 2 - Soft tissue involvement

- First stage of infiltrative ophthalmopathy
- Symptoms
  - Lacrimation
  - Photophobia
  - FB sensation
  - Retrobulbar discomfort



Marcocci, C., Smith, T.J. (2016). Graves' Ophthalmopathy. In: Vitti, P., Hegedus, L. (eds) Thyroid Diseases. Endocrinology. Springer, Cham. https://doi.org/10.1007/978-3-319-29195-6_16-1

NOSPECS

Class 2 - Soft tissue involvement

- Signs
  - Lid/conjunctival edema
  - Lid/conjunctival hyperemia
  - Conjunctival chemosis
  - Extrusion of orbital fat
  - Lacrimal gland swelling
  - Inflammation near EOM insertion



- Management:
  - Lubricants, taping lids, oculoplastic consult

Marcocci, C., Smith, T.J. (2016). Graves' Ophthalmopathy. In: Vitti, P., Hegedus, L. (eds) Thyroid Diseases. Endocrinology. Springer, Cham. https://doi.org/10.1007/978-3-319-29195-6_16-1

NOSPECS

Class 3 - Proptosis (aka “exophthalmos”)

- Hyperthyroidism is most common cause of unilateral proptosis
- 80% of those patients with bilateral exophthalmos will have Graves' disease
- Management:
  - Lubricants
  - Consult with endocrinologist



Marcocci, C., Smith, T.J. (2016). Graves’ Ophthalmopathy. In: Vitti, P., Hegedus, L. (eds) Thyroid Diseases. Endocrinology. Springer, Cham. https://doi.org/10.1007/978-3-319-29195-6_16-1

NOSPECS

- Class 4 - EOM involvement
  - Palsy → diplopia
    - IR most commonly affected muscle (60-70%) followed by MR
    - IOP elevation in upgaze
      - ~20%
      - Secondary to tractional effect



- Management:
  - Lubricants
  - Patch until thyroid condition under control

Marcocci, C., Smith, T.J. (2016). Graves' Ophthalmopathy. In: Vitti, P., Hegedus, L. (eds) Thyroid Diseases. Endocrinology. Springer, Cham. https://doi.org/10.1007/978-3-319-29195-6_16-1

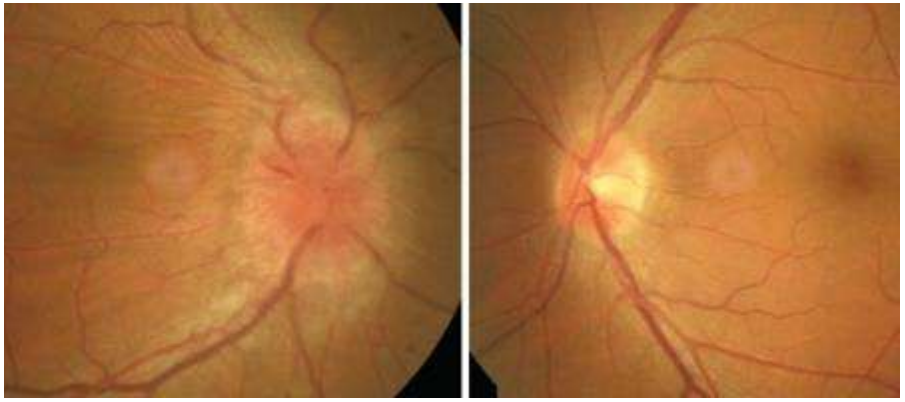
NOSPECS

- Class 5 - Corneal involvement
  - Exposure keratopathy
    - Can pose serious threat to vision
  - Management:
    - Lubes (ung = ointment)
    - Taping
    - Steroids topical and or oral



NOSPECS

- Class 6 - Sight loss (ON involvement)
  - Optic disc edema
    - Painless, gradual loss of vision
  - Management:
    - Lubricants
    - IV steroids
    - Surgical Decompression
    - NOTHING???→ Remember- 70% improve without Tx!



Wang, AG. (2018). Thyroid Eye Disease with Compressive Optic Neuropathy. In: Emergency Neuro-ophthalmology . Springer, Singapore. https://doi.org/10.1007/978-981-10-7668-8_28

Classification: Clinical Activity Score (CAS)

- Assess treatment of active inflammatory ophthalmopathy
- Predict therapeutic outcome, and
- Select patients for surgical or non-surgical treatment

- High predictive outcome for:
  - Immunosuppressive treatment for infiltrative ophthalmopathy**

Clinical Activity Score (CAS)

Proposed Classification System to Assess Disease Activity in TED

- One point is given for each sign present. The sum of these points defines activity score (From Mourits et al.)
  - **Pain**
  - **Painful, oppressive feeling on or behind the globe**
  - **Pain on attempted up, side, or down gaze**
  - **Eyelid redness**
  - **Eyelid edema**
  - **Diffuse conjunctival redness**
  - **Chemosis (conjunctiva edema)**
  - **Increase proptosis of 2 mm or more between 1 and 3 months**
  - **Impaired function**
  - **Decrease in visual acuity of 1 or more lines on Snellen chart (using pinhole) between 1 and 3 months**
  - **Decrease of eye movements in any direction equal to or more than 5 degrees between 1 and 3 months**

- Score ≥ 4
  - Positive predictive value 80%
  - Negative predictive value 64%

Adapted from http://www.thyroidmanager.org/Chapter12/ch01s02.html

Classification: EUGOGO

http://www.eugogo.eu/

European Group on Graves Orbitopathy

1. **Sight-threatening** due to optic neuropathy or corneal breakdown
2. **Moderate-severe** with one of moderate-severe soft tissue involvement, lid retraction 2mm or more, diplopia and proptosis 3mm or more
3. **Mild** with only minor impact on daily life

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EUGOGO

Suggests calculating clinical activity score (CAS) to help determine threshold for use of immunosuppressives

- Score of 3 or more – consider treatment

EUGOGO also use NOSPECS to determine treatment plans

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EUGOGO

CAS for EUGOGO:

- Spontaneous orbital pain
- Gaze evoked orbital pain
- Eyelid swelling considered due to active (inflammatory phase) TED
- Eyelid erythema
- Conjunctival redness considered due to active (inflammatory phase) TED
- Chemosis
- Inflammation of caruncle or plica
- Increase in proptosis of 2mm or more,
- Decrease in uniocular excursion in any one direction of 8° or more, or decrease in Snellen acuity of one line

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Classification- VISA

- Vision loss
- Inflammation/congestion and activity in TED
- Strabismus / motility
- Appearance / exposure

http://thyroideyedisease.org/clinical-features-visa-classification/

Thyroid Eye Disease

Optometric work-up:

- Best corrected visual acuities
- Color vision
- Pupils
- Extraocular motilities
- Lid fissure height in primary gaze
- Exophthalmometry
- Slit lamp
- Dilated fundus exam
- Imaging (photos, OCT) if needed

Thyroid Eye Disease

Blood Tests :

- Thyroid Panel to include:
  - Thyroid Stimulating Hormone immunoradiometric assay (TSH-IRMA, sensitive TSH (sTSH))
  - Thyroid Stimulating Immunoglobulin (TSI)
  - T3 and T4
- T3>>T4 → Think Graves
- T3 & T4 normal with abnormal TSH →Euthyroid

Thyroid Eye Disease

Imaging

- External and Fundus Photos, OCT
- B-scan ultrasound
- CT scan
- MRI

Thyroid Eye Disease

Medical Management

- Correct thyroid dysfunction
  - Work with endocrinologist
- Treat the TED
  - Ocular lubricants
  - Topical steroids and/or antibiotics
  - TEPEZZA

Thyroid Eye Disease

Medical Management

- Treating the TED (continued)
- Systemic Steroids mainstay treatment for moderate to severe disease (severe corneal involvement)
  - Oral prednisolone 60 – 80 mg/day
  - Tapered
- IV methylprednisolone reserved for compressive optic neuropathy

Duane's Ophthalomology – Chapter 36 – Thyroid Orbitopathy
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Thyroid Eye Disease

- **Surgical Management**
  - Surgical decompression
    - One or two holes made in orbit bones
      - Most common: medial wall and floor
  - Surgical treatment of EOM abnormalities when stable
- **Radiation Management**

Medical and Surgical Management of Graves' Ophthalmopathy

Symptom/Sign	*Management*
Eye discomfort (e.g., dryness, "gritty sensation") and eyelid retraction	Ocular lubricants Eyelids closed with adhesive tape during sleep Dark spectacle lenses Adrenergic blocking agents (e.g., guanethidine) Eyelid surgery
Periorbital edema, chemosis, injection	Diuretics Sleep with head of bed elevated Corticosteroids Immunosuppressants Radiotherapy I.V. immunoglobulin Plasmapheresis Orbital decompression surgery
Diplopia	Patching or lens occlusion Prism eyeglasses Extraocular muscle surgery
Disfiguring proptosis	Orbital decompression Eyelid surgery
Decreased visual acuity (i.e., optic nerve compression)	Corticosteroids Immunosuppressants Radiotherapy Orbital decompression

Adapted from Garrity JA. Graves' ophthalmopathy: An ophthalmologist's perspective. Thyroid Today 1992; 15 (No. 1): 1–9 and Bahn RS, Garrity JA, Gorman CA. Diagnosis and management of Graves' ophthalmopathy. J. Clin. Endocrinol Metab 1990; 71:559–563.

CLINICAL COURSE

- Variable
  - Ocular involvement inevitable
  - Intermittent
  - Proptosis not necessarily affected by thyroid TX

- Onset → remission ~ 6 months to 3 years

THANK YOU!